

## Introduction :

In this project, I have two files of data given by a local government's department of transport which reveal the travel habits of its citizens. I should analyze these data by using my knowledge of probability models in engineering.

## Question 1

a) It isn't a math question. It's only about selecting data from MATLAB. So I used MATLAB to extract and keep only trips have MODE = 3, MODE = 4 or MODE = 6 (trips by public transport).

```
clear
close all;
% DATA_Trip_File = xlsread("DATA_Trip_File.csv");
load('DATA_Trip_File.mat')

Position_3 = find(DATA_Trip_File(:,11)==3);% MODE=3
Position_4 = find(DATA_Trip_File(:,11)==4);% MODE=4
Position_6 = find(DATA_Trip_File(:,11)==6);% MODE=6
Position = [Position_3;Position_4;Position_6];

DATA_Trip_File_Public = DATA_Trip_File(Position,:);
```

I got a new table named DATA\_Trip\_File\_Public and will use this table in rest of the Question 1.

|    | 1        | 2 | 3  | 4 | 5 | 6 | 7    | 8      | 9   | 10 | 11 | 12 | 13     | 14 |
|----|----------|---|----|---|---|---|------|--------|-----|----|----|----|--------|----|
| 1  | 13021522 | 1 | 53 | 2 | 2 | 1 | 1500 | 10     | 240 | 4  | 3  | 4  | 0.9000 |    |
| 2  | 13017622 | 1 | 53 | 2 | 2 | 1 | 5230 | 10     | 30  | 4  | 3  | 5  | 0.9000 |    |
| 3  | 13021522 | 1 | 53 | 2 | 2 | 1 | 720  | 10     | 175 | 4  | 3  | 4  | 0.9000 |    |
| 4  | 13017877 | 2 | 20 | 2 | 2 | 2 | 1700 | 4      | 40  | 5  | 3  | 4  | 0.9000 |    |
| 5  | 13021877 | 2 | 20 | 2 | 2 | 2 | 800  | 4      | 40  | 5  | 3  | 4  | 0.9000 |    |
| 6  | 13017791 | 1 | 49 | 1 | 2 | 2 | 1135 | 4      | 25  | 6  | 3  | 4  | 0.9000 |    |
| 7  | 13062402 | 2 | 15 | 1 | 2 | 2 | 5000 | 4      | 15  | 3  | 3  | 2  | 0.4000 |    |
| 8  | 13062414 | 1 | 38 | 2 | 1 | 1 | 1700 | 14     | 80  | 4  | 3  | 4  | 1      |    |
| 9  | 1305214  | 1 | 38 | 2 | 1 | 1 | 1545 | 14     | 45  | 4  | 3  | 5  | 15     |    |
| 10 | 1305214  | 2 | 12 | 2 | 2 | 2 | 1700 | 14     | 80  | 4  | 3  | 4  | 15     |    |
| 11 | 1306214  | 2 | 12 | 2 | 2 | 2 | 1545 | 14     | 45  | 4  | 3  | 4  | 15     |    |
| 12 | 1306152  | 2 | 25 | 2 | 2 | 2 | 1430 | 0.5600 | 10  | 3  | 3  | 1  | 0.1000 |    |
| 13 | 1306152  | 2 | 25 | 2 | 2 | 2 | 1105 | 0.1200 | 5   | 3  | 3  | 5  | 0.0200 |    |

b). I should sift through TRPMILES (the distance of the particular trip) and R-AGE (for people from the same household, consider a person's age only once) from the previous table firstly.

```
close all;
Same_HouseID = diff(DATA_Trip_File_Public(:,1));
Same_PersonID = diff(DATA_Trip_File_Public(:,2));
Same_Person = (Same_HouseID == 0) & (Same_PersonID == 0);
Repetition_RowNum = find(Same_Person == 1)+1;
DATA_Trip_File_Public_Q1b = DATA_Trip_File_Public;
DATA_Trip_File_Public_Q1b(Repetition_RowNum,:) = [];

Age_Q1b = DATA_Trip_File_Public_Q1b(:,3);
TripMiles_Q1b = DATA_Trip_File_Public(:,8);
```

← if HomeID is the same,  
the personID cannot be the  
same  
marked and added up.

Two new tables:

| Age_Q1b | TripMiles_Q1b |
|---------|---------------|
| 1       | 2             |
| 53      |               |
| 20      |               |
| 69      |               |
| 15      |               |
| 38      |               |
| 12      |               |
| 25      |               |
| 11      |               |
| 33      |               |
| 49      |               |
| 18      |               |
| 10      |               |

| Age_Q1b | TripMiles_Q1b |
|---------|---------------|
| 1       | 2             |
| 10      |               |
| 10      |               |
| 10      |               |
| 4       |               |
| 4       |               |
| 4       |               |
| 14      |               |
| 14      |               |
| 14      |               |
| 14      |               |
| 0.5600  |               |
| 0.3300  |               |

Then, I analysed these two kind of data separately and determined the most appropriate probability distribution to model each of the two random variables. And as mentioned, I should consider each of them as continuous.

For continuous random variables, here are some common probability distributions:

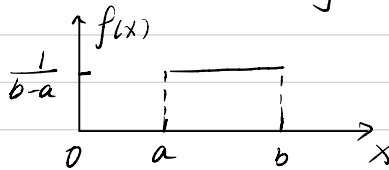
1. uniform distribution.
2. Gaussian distribution.
3. Exponential distribution.
4. Gamma distribution.

Uniform distribution :

if continuous random variables have probability density function like that :

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise,} \end{cases}$$

it obeys uniform distribution. In layman's terms, it means these random variables only have two kinds of PPTs.

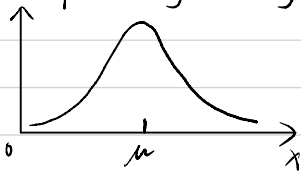


It is impossible for the random variables from R-AGE or TRPMILES which are irregular, stochastic. The most common example of this distribution is tossing coin.

Gaussian distribution :

if continuous random variable satisfy  $X = N(\mu, \sigma^2)$ , it obeys Gaussian distribution. ( $\mu$  is expectation,  $\sigma^2$  is standard deviation)

Its probability density function is  $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ .



The characteristic of its graph:

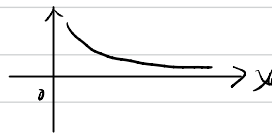
1. total Area equal to 1
2. symmetry center is  $\mu$
3. the bigger  $\sigma$ , the more smooth curve.

R-AGE and TRPMILES might satisfy this distribution.

Exponential distribution:

The biggest difference between exponential distribution and other distributions is that it only refers to probabilities of time interval of the event.

PDF:  $f_x(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

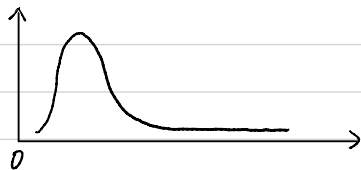


I don't think R\_AGE or TRPMILES will satisfy this model, but I still double checked as blew.

Gamma distribution:

It seems like the sum of several exponential independent random variables.

PDF:  $f_x(x) = \frac{\lambda (\lambda x)^{a-1} e^{-\lambda x}}{\Gamma(a)}, \quad x \in (0, +\infty)$

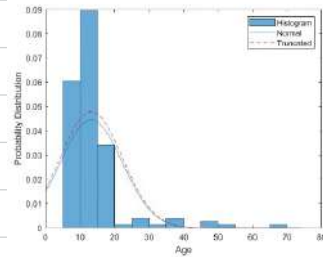


After a elementary expectation, I used MATLAB to draw the graph to observe which distribution is the most suitable.



For R-AGE :

Normal distribution  
 $\mu = 13.2434$   
 $\sigma = 8.95389$   
 Truncated to the interval [0, 69]



nll normal = 548.3769

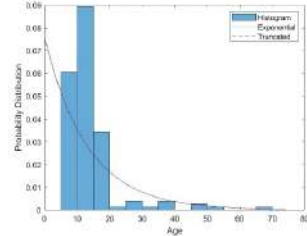
```
% Try Normal Distribution
pd_normal = fitdist(Age_Q1b, 'Normal')
figure('Name', 'Histogram and Fitted Normal Distribution');
histogram(Age_Q1b, 'Normalization', 'pdf');
pd_normal_x_axis = 0:0.1:max(Age_Q1b)+5;
pd_normal_y_axis = pdf(pd_normal, pd_normal_x_axis);
line(pd_normal_x_axis, pd_normal_y_axis)
t_normal = truncate(pd_normal, 0, max(Age_Q1b))
line(pd_normal_x_axis, pdf(t_normal, pd_normal_x_axis), 'LineStyle', 'dashed')
legend('Histogram', 'Normal', 'Truncated')
xlabel('Age')
ylabel('Probability Distribution')
nll_normal = negloglik(pd_normal)
```

`fitdist(X, distname)` : creates a probability distribution object by fitting the distribution specified by `distname` to the data in column vector `X`.

`truncate(pd, lower, upper)` .

```
% Try Exponential Distribution
pd_exp = fitdist(Age_Q1b, 'Exponential')
figure('Name', 'Histogram and Fitted Exponential Distribution');
histogram(Age_Q1b, 'Normalization', 'pdf');
pd_exp_x_axis = 0:0.1:max(Age_Q1b)+5;
pd_exp_y_axis = pdf(pd_exp, pd_exp_x_axis);
line(pd_exp_x_axis, pd_exp_y_axis)
t_exp = truncate(pd_exp, 0, max(Age_Q1b))
line(pd_exp_x_axis, pdf(t_exp, pd_exp_x_axis), 'LineStyle', '--', 'r')
legend('Histogram', 'Exponential', 'Truncated')
xlabel('Age')
ylabel('Probability Distribution')
nll_exp = negloglik(pd_exp)
```

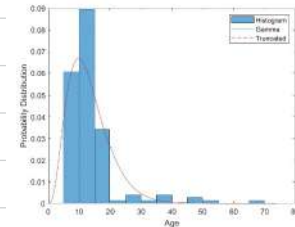
Exponential distribution  
 $\mu = 13.2434$   
 Truncated to the interval [0, 69]



nll exp = 544.6921

```
% Try Gamma Distribution
pd_Gamma = fitdist(Age_Q1b, 'Gamma')
figure('Name', 'Histogram and Fitted Gamma Distribution');
histogram(Age_Q1b, 'Normalization', 'pdf');
pd_Gamma_x_axis = 0:0.1:max(Age_Q1b)+5;
pd_Gamma_y_axis = pdf(pd_Gamma, pd_Gamma_x_axis);
line(pd_Gamma_x_axis, pd_Gamma_y_axis)
t_Gamma = truncate(pd_Gamma, 0, max(Age_Q1b))
line(pd_Gamma_x_axis, pdf(t_Gamma, pd_Gamma_x_axis), 'LineStyle', 'dashed', 'r')
legend('Histogram', 'Gamma', 'Truncated')
xlabel('Age')
ylabel('Probability Distribution')
nll_Gamma = negloglik(pd_Gamma)
```

Gamma distribution  
 $a = 3.87216$   
 $b = 3.42017$   
 Truncated to the interval [0, 69]



nll gamma = 491.5285

These reveals that

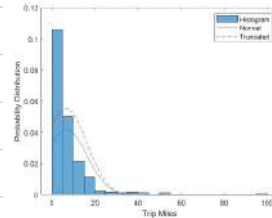
$nll\_Gamma < nll\_exp < nll\_normal$ , so the most appropriate distribution for R-AGE is Gamma distribution.

## For TRIMILES:

% For the variable of TRIMILE

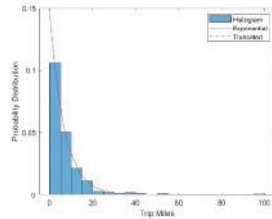
```
% Try Normal Distribution
pd_normal = fitdist(TripMiles_Q1b,'Normal');
figure('Name','Histogram and Fitted Normal Distribution');
histogram(TripMiles_Q1b,'Normalization','pdf');
pd_normal_x_axis = 0:0.1:max(TripMiles_Q1b)+5;
pd_normal_y_axis = pdf(pd_normal,pd_normal_x_axis);
line(pd_normal_x_axis,pd_normal_y_axis)
t_normal = truncate(pd_normal,0,max(TripMiles_Q1b))
line(pd_normal_x_axis,pdf(t_normal, pd_normal_x_axis),'LineStyle','--')
legend('Histogram','Normal','Truncated')
xlabel('Trip Miles')
ylabel('Probability Distribution')
nll_normal = negloglik(pd_normal)
```

Normal distribution  
mu = 6.69326  
sigma = 9.51115  
Truncated to the interval [0, 100]



nll\_normal = 1.0679e+03

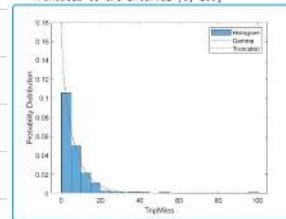
Exponential distribution  
mu = 6.69326  
Truncated to the interval [0, 100]



nll\_exp = 844.2206

```
% Try Gamma Distribution
pd_Gamma = fitdist(TripMiles_Q1b,'Gamma');
figure('Name','Histogram and Fitted Gamma Distribution');
histogram(TripMiles_Q1b,'Normalization','pdf');
pd_Gamma_x_axis = 0:0.1:max(TripMiles_Q1b)+5;
pd_Gamma_y_axis = pdf(pd_Gamma,pd_Gamma_x_axis);
line(pd_Gamma_x_axis,pd_Gamma_y_axis)
t_Gamma = truncate(pd_Gamma,0,max(TripMiles_Q1b))
line(pd_Gamma_x_axis,pdf(t_Gamma, pd_Gamma_x_axis),'LineStyle','--')
legend('Histogram','Gamma','Truncated')
xlabel('Trip Miles')
ylabel('Probability Distribution')
nll_Gamma = negloglik(pd_Gamma)
```

Gamma distribution  
a = 0.9308953  
b = 7.1063d  
Truncated to the interval [0, 100]



nll\_gamma = 843.7113

The most appropriate probability distribution for TRPMILES is Gamma distribution.

c) For this part, I should calculate the average TRPMILES per person firstly.  
`Same_Person_Completion = [0; Same_Person; 0];` → 0 are added to the front and back of  
`DATA_Trip_File_Public_Copy = DATA_Trip_File_Public;` Same-Person before using diff.

`k = diff(Same_Person_Completion);`  
`ind = find(k==1)+1;`  
`num = find(k==1)-ind+1;`  
`clear Average_TripMiles`

They are used to ensure that the first and last element is 1, the -1 or 1 can be found by diff and then find position and number based on them.

selecting data still a little bit complex.

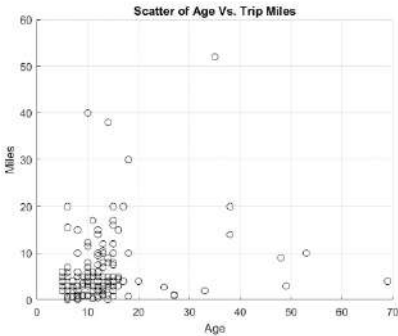
```
for n = 1:length(ind)
    Person_Num = [ind(n)-1:ind(n)+num(n)-1];
    Average_Age(n) = mean(DATA_Trip_File_Public(Person_Num,3));
    Average_TripMiles(n) = mean(DATA_Trip_File_Public(Person_Num,8));
    DATA_Trip_File_Public_Copy(Person_Num,:) = zeros;
end

Zero_Row_Num = find(DATA_Trip_File_Public_Copy(:,1)==0);
DATA_Trip_File_Public_Copy(Zero_Row_Num,:) = [];

Age_Q1c = [Average_Age, DATA_Trip_File_Public_Copy(:,3)'];
Average_TripMiles_Q1c = [Average_TripMiles, DATA_Trip_File_Public_Copy(:,8)']
```

Plotting the scatter

```
close all
figure(3)
scatter(Age_Q1c, Average_TripMiles_Q1c, 'k')
title('Scatter of Age Vs. Trip Miles')
grid on
xlabel('Age')
ylabel('Miles')
% hold on
% scatter(DATA_Trip_File_Public_Copy(:,3), DATA_Trip_File_Public_Copy(:,8), 'k')
```



Through observing this average miles vs. ages scatter, I suspect that NOT a linear relationship between these two variables.

Now, I should consider "Sample correlation coefficient". Its formula is

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{VAR}(X) \text{VAR}(Y)}}$$

$\text{cov}(X, Y)$  the covariance,  $\text{VAR}$  are variances.  $r \in [-1, 1]$   
 when  $r$  is close to  $\pm 1$ , it indicates that the data has a strong linear correlation; when  $r$  is close to 0, it indicates that data has no linear correlation.

Based on its properties above, I predicted the sample correlation coefficient is close to zero because I don't observe any linear relationship of two variables from scatter.

d) The formula:  $\text{Cov}(X, Y) = E\{[X - E(X)][Y - E(Y)]\}$ .  
 I used four methods to compute the correlation coefficient.

```
% Method I
Correlation_Coefficient = corrcoef(Age_Q1c, Average_TripMiles_Q1c)

% Method II
Covariance_2 = cov(Age_Q1c, Average_TripMiles_Q1c);
Correlation_Coefficient_2 = Covariance_2/(std(Age_Q1c)*std(Average_TripMiles_Q1c))

% Method III
E_Age = mean(Age_Q1c);
E_Miles = mean(Average_TripMiles_Q1c);
Covariance_3 = mean((Age_Q1c - E_Age).*(Average_TripMiles_Q1c - E_Miles));
Correlation_Coefficient_3 = Covariance_3/(std(Age_Q1c)*std(Average_TripMiles_Q1c))

% Method IV
n = length(Age_Q1c);
xBar = mean(Age_Q1c);
yBar = mean(Average_TripMiles_Q1c);
numerator = 0;
for i = 1:n
    numerator = numerator + (Age_Q1c(i) - xBar)*(Average_TripMiles_Q1c(i) - yBar);
end
Covariance_4 = numerator/(n-1); % see https://au.mathworks.com/help/matlab/ref/cov.html
Correlation_Coefficient_4 = Covariance_4/(std(Age_Q1c)*std(Average_TripMiles_Q1c))
```

```
Correlation_Coefficient = 2x2
    1.0000    0.1947
    0.1947    1.0000
```

```
Correlation_Coefficient_2 = 2x2
    1.2150    0.1947
    0.1947    0.8231
```

```
Correlation_Coefficient_3 =
    0.1934
```

```
Correlation_Coefficient_4 = 0.1947
```

Although I can compute coefficient by MATLAB in different ways,

the most straightforward way is using the mathematic formula. After all, the sample correlation coefficient is 0.1947. This value is close to my prediction, and it reveals that NO Linear relationship.

## Question 2.

a) The aim of this question is construct a joint PMF for TOTAL and HH SIZE. Consider TOTAL and HH SIZE are discrete random variables.

Joint PMF means a PMF has several variables. For two variables, it can be defined:

$$P_{X,Y}(x,y) = P(X=x, Y=y)$$

or I can write:

$$P_{X,Y}(x,y) = P((X=x) \text{ and } (Y=y))$$

and the sum of probabilities is equal to one:

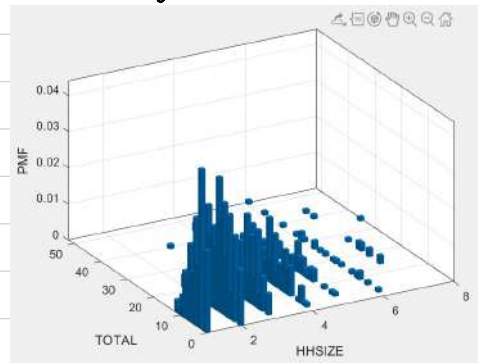
$$\sum_x \sum_y P_{X,Y}(x,y) = 1$$

I wrote down a MATLAB function to construct joint PMF of these cases.

```
function PMF_T_H = PMF(HH_SIZE_Value, TOTAL_Value, TOTAL, HH_SIZE)
Overall_Case = length(TOTAL);
TOTAL_Case = find(TOTAL == TOTAL_Value);
HH_SIZE_Case = find(HH_SIZE == HH_SIZE_Value);
Target_Case = intersect(TOTAL_Case, HH_SIZE_Case);
PMF_T_H = length(Target_Case)/Overall_Case;
end
```

calculate the probabilities and then use the definition.

we can observe the joint PMF in a intuitive way:



or observe them in the table 'PMF\_HHSIZE\_TOTAL' :

|    | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     | 12     | 13     | 14 |
|----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|----|
| 1  | 0.0227 | 0.0012 | 0.0442 | 0.0179 | 0.0296 | 0.0143 | 0.0215 | 0.0119 | 0.0131 | 0.0060 | 0.0034 | 0.0036 | 0      | 0  |
| 2  | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 3  | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 4  | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 5  | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 6  | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 7  | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 8  | 0.0155 | 0      | 0.0239 | 0.0384 | 0.0286 | 0.0167 | 0.0310 | 0.0263 | 0.0170 | 0.0179 | 0.0227 | 0.0191 | 0.0274 | 0  |
| 9  | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 10 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 11 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 12 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 13 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 14 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 15 | 0      | 0      | 0.0060 | 0      | 0.0095 | 0.0048 | 0.0215 | 0.0084 | 0.0191 | 0.0064 | 0.0239 | 0.0064 | 0.0155 | 0  |
| 16 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 17 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 18 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 19 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 20 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |
| 21 | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0  |

b) I should compute  $P(\text{HHSIZE}=3, \text{TOTAL}=4)$ ,  $P(\text{HHSIZE}=3, \text{TOTAL} \leq 4)$ ,  $P(\text{HHSIZE} \leq 3, \text{TOTAL} \leq 4)$  in this part. Because of the previous question a, I just need sift some criteria from the numbers I have already obtained.

(i)  $P(\text{HHSIZE} = 3, \text{TOTAL} = 4)$

$$\therefore P(\text{HHSIZE}=3, \text{TOTAL}=4) = 0.0095$$

```
PMF_H3_T4 = PMF(3,4,TOTAL,HHSIZE)
```

(ii)  $P(\text{HHSIZE} = 3, \text{TOTAL} \leq 4)$

$$\therefore P(\text{HHSIZE}=3, \text{TOTAL} \leq 4) = 0.0155$$

```
PMF_H3_TLessThan4 = 0
for n = 0:4
    PMF_H3_TLessThan4 = PMF_H3_TLessThan4 + PMF(3,n,TOTAL,HHSIZE)
end
```

(iii)  $P(\text{HHSIZE} \leq 3, \text{TOTAL} \leq 4)$

```
PMF_HLessThan3_TLessThan4 = 0
for n = 1:3
    for k = 0:4
        PMF_HLessThan3_TLessThan4 = PMF_HLessThan3_TLessThan4 + PMF(n,k,TOTAL,HHSIZE)
    end
end
```

$$\therefore P(\text{HHSIZE} \leq 3, \text{TOTAL} \leq 4) = 0.2076$$

c) I need to find marginal PMFs for HHSIZE and TOTAL. Definition of marginal probability is that 'when we know the joint probability of a set of variables, we want to know the probability of a subset of them.' For discrete-time variables,

$$P(X=x) = \sum_y P(X=x, Y=y)$$

The Marginal PMF for HHSIZE

```
function PMF_Marginal_HHSIZE = PMF_Marginal_H(HHSIZE_Value, TOTAL, HHSIZE)
% Sort the TOTAL Array
TOTAL_Sort = sort(TOTAL,1,"ascend");
Diff_Total = diff(TOTAL_Sort);
Repetition_Num_TOTAL = find(Diff_Total==0)+1;
TOTAL_Sort(Repetition_Num_TOTAL,:) = [];

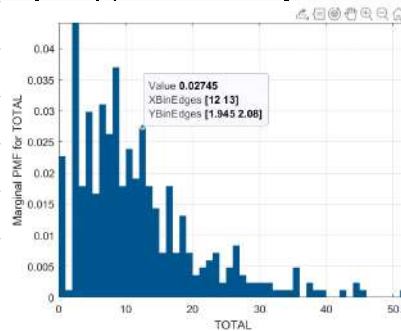
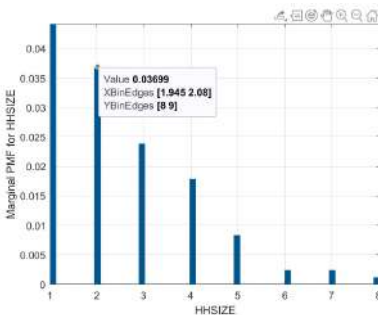
PMF_Marginal_HHSIZE = 0;
for n = 1:length(TOTAL_Sort)
    PMF_Marginal_HHSIZE = PMF_Marginal_HHSIZE + PMF(HHSIZE_Value, TOTAL_Sort(n), TOTAL, HHSIZE);
end
end
```

The Marginal PMF for TOTAL

```
function PMF_Marginal_TOTAL = PMF_Marginal_T(TOTAL_Value, TOTAL, HHSIZE)
% Sort the HHSIZE Array
HHSIZE_Sort = sort(HHSIZE,1,"ascend");
Diff_HHSIZE = diff(HHSIZE_Sort);
Repetition_Num_HHSIZE = find(Diff_HHSIZE==0)+1;
HHSIZE_Sort(Repetition_Num_HHSIZE,:) = [];

PMF_Marginal_TOTAL = 0;
for n = 1:length(HHSIZE_Sort)
    PMF_Marginal_TOTAL = PMF_Marginal_TOTAL + PMF(HHSIZE_Sort(n), TOTAL_Value, TOTAL, HHSIZE);
end
end
```

As the result, the marginal PMFs for HHSIZE and for TOTAL are as below:





we can get specific values by placing our mouse over the top of each bar.

d) In this part, I should consider the conditional probability.  
 Definition: the probability of A given knowledge that the event B has occurred is referred to as the conditional probability of A given B,

$$P(A|B) = \frac{P(A, B)}{P(B)}, \quad P(B) \neq 0$$

$$\text{So } P(\text{TOTAL}=4 | \text{HHSIZE}=3) = \frac{P(\text{TOTAL}=4, \text{HHSIZE}=3)}{P(\text{HHSIZE}=3)}$$

and so on ...

For MATLAB code, the previous pares for Joint and marginal are pared for this.

(i)  $P(\text{TOTAL} = 4 | \text{HHSIZE} = 3);$

```
PMF_T4_H3 = PMF(3,4,TOTAL,HHSIZE)
PMF_H3 = PMF_Marginal_H(3,TOTAL,HHSIZE)
P_T4_WhenH3 = PMF_T4_H3/PMF_H3
```

→ numerator

→ denominator (focus on one subset, so it can be calculated by marginal PMF.)

(ii)  $P(\text{HHSIZE} = 3 | \text{TOTAL} = 4);$

```
PMF_T4_H3 = PMF(3,4,TOTAL,HHSIZE)
PMF_T4 = PMF_Marginal_T(4,TOTAL,HHSIZE)
P_H3_WhenT4 = PMF_T4_H3/PMF_T4
```

(iii)  $P(\text{HHSIZE} = 3 | 2 \leq \text{TOTAL} \leq 4)$

```
PMF_H3_TFrom2to4 = PMF(3,2,TOTAL,HHSIZE)+PMF(3,3,TOTAL,HHSIZE)+PMF(3,4,TOTAL,HHSIZE)
PMF_TFrom2to4 = PMF_Marginal_T(2,TOTAL,HHSIZE)+PMF_Marginal_T(3,TOTAL,HHSIZE)+PMF_Marginal_T(4,TOTAL,HHSIZE)
PMF_H3_When_TFrom2to4 = PMF_H3_TFrom2to4/PMF_TFrom2to4
```

Answer:

(i)  $p = 0.0485$

(ii)  $p = 0.1290$

(iii)  $p = 0.0336$



e) In this part I need to determine whether TOTAL and HH SIZE are independent random variables or not.  
 For random variables  $x, y$ 's cumulative distribution function is  $F(x, y)$ ,  $F_x(x)$  and  $F_y(y)$  are marginal distribution function, if  $F(x, y) = F_x(x) F_y(y)$ ,  
 so  $x, y$  are independent.

For TOTAL, HH SIZE, MATLAB code as:

```
PMF_H3_T4 = PMF(3,4,TOTAL,HH SIZE)
PMF_T4 = PMF_Marginal_T(4,TOTAL,HH SIZE)
PMF_H3 = PMF_Marginal_H(3,TOTAL,HH SIZE)
A = PMF_T4*PMF_H3
```

results:

PMF\_T4\_H3 = 0.0095

A = 0.0146

PMF\_T4\_H3  $\neq$  PMF\_T4 \* PMF\_H3

So TOTAL and HH SIZE are dependent random variables.

### Question 3.

a) Consider DATA\_Trip\_File.csv file and only work with data that relates to trips made by public transport as question 1. Compute the average TRPMILES that a person made. Or in other words, compute person A from a household made several trips, the average tripmiles is the average distance of all the trips by A. To do this, I wrote the following code:

```
% Extract the data which related to public transport
Position_3 = find(DATA_Trip_File(:,11)==3); % MODE=3
Position_4 = find(DATA_Trip_File(:,11)==4); % MODE=4
Position_6 = find(DATA_Trip_File(:,11)==6); % MODE=6
Position = [Position_3; Position_4; Position_6];

DATA_Trip_File_Public = DATA_Trip_File(Position,:);
```

Firstly, sift data related to public transport as Question 1.

```

Same_HouseID = diff(DATA_Trip_File_Public(:,1)); % Check if adjacent House is the same House
Same_PersonID = diff(DATA_Trip_File_Public(:,2)); % Check if adjacent Person shares the same Person ID
% Check if adjacent Person is the same Person
% Same_Person = 1 --> the same person; Same_Person = 0 --> not the same person;
Same_Person = (Same_HouseID == 0) & (Same_PersonID == 0);

Same_Person_Completion = [0; Same_Person; 0]; % Add 0 before and after Same_Person and then use diff,

% Copy the original data
DATA_Trip_File_Public_Copy = DATA_Trip_File_Public;

k = diff(Same_Person_Completion);
% add 1 is for the right index after diff()
% ind denotes the
ind = find(k==1)+1;
% the position differences between -1 and 1 appears is the number of times of 1
num = find(k==1)-ind+1;

% clear Average_TripMiles

for n = 1:length(ind)
    Person_Num = [ind(n)-1:ind(n)+num(n)-1]; % repeatedly persons' index (no. n group), react original data's row
    Average_Age(n) = mean(DATA_Trip_File_Public(Person_Num,3));
    Average_TripMiles(n) = mean(DATA_Trip_File_Public(Person_Num,8));
    % make data from DATA_Trip_File_Public_Copy which already 'averaged' equal to zero, delete after the loop
    DATA_Trip_File_Public_Copy(Person_Num,:) = zeros;
end

% find the rows from DATA_Trip_File_Public_Copy which were made in zero
Zero_Row_Num = find(DATA_Trip_File_Public_Copy(:,1)==0);
% delete them, finally i got good samples
DATA_Trip_File_Public_Copy(Zero_Row_Num,:) = [];

% completely data
Age_Needed = [Average_Age, DATA_Trip_File_Public_Copy(:,3)'];
Average_TripMiles_Needed = [Average_TripMiles, DATA_Trip_File_Public_Copy(:,8)'];

```

Secondly, using a very long piece of code to calculate the average TRPMILES per person. For details, see MATLAB code and the green explanation. After that, I got a table of average TRPMILES.

| Average_TripMiles_Needed |   |    |    |        |    |   |    |    |    |    |    |    |    |  |
|--------------------------|---|----|----|--------|----|---|----|----|----|----|----|----|----|--|
| 1x152 double             |   |    |    |        |    |   |    |    |    |    |    |    |    |  |
| 1                        | 2 | 3  | 4  | 5      | 6  | 7 | 8  | 9  | 10 | 11 | 12 | 13 | 14 |  |
| 10                       | 4 | 14 | 14 | 2.7225 | 17 | 3 | 30 | 10 | 40 | 10 | 15 | 2  |    |  |

Next, Let  $S_n \rightarrow$  sum of all these average TRPMILE measurements,  
 $M_n \rightarrow$  sample mean of the average TRPMILE measurements,  
 $n \rightarrow$  total number of measurements.

then compute  $M_n$ .

Sample mean is a pretty straightforward thing:

Let  $X_i$ s are IID random variables with the same PDF as  $X$ ,  $X$  be a random variables with  $E[X] = m$ , the sample mean is

$$M_n = \frac{1}{n} \sum_{j=1}^n X_j = \frac{S_n}{n}$$

So it is easy to compute  $M_n$ :

a) Compute  $M_n$ .

```
Sn(1) = Average_TripMiles_Needed(1)% Initialized the sum of n samples
Mn(1) = Sn(1)/1

for n=2:length(Average_TripMiles_Needed)
    Sn(n) = Sn(n-1) + Average_TripMiles_Needed(n);
    Mn(n) = Sn(n)/n;
end

close all;

n = length(Average_TripMiles_Needed);

figure(2)
plot(1:n,Sn)
title('Sn Vs. n')

figure(3)
plot(1:n,Mn)
title('Mn Vs. n')
```

$$\underline{M_n = 6.2705}$$

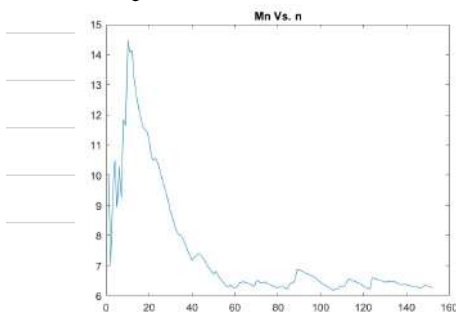
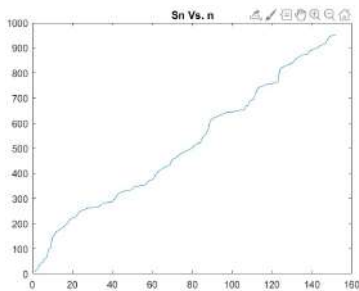
| Mn           | 1  | 2 | 3      | 4       | 5      | 6       | 7      | 8       | 9       | 10      | 11      | 12      | 13      | 14    |
|--------------|----|---|--------|---------|--------|---------|--------|---------|---------|---------|---------|---------|---------|-------|
| 1x152 double | 10 | 7 | 9.3333 | 10.5000 | 8.9445 | 10.2871 | 9.2461 | 11.6403 | 11.6358 | 14.4722 | 14.0657 | 14.1435 | 13.2094 | 12.68 |

$M_n$  table

| Mn           | 1  | 2  | 3  | 4  | 5       | 6       | 7       | 8       | 9        | 10       | 11       | 12       | 13       | 14    |
|--------------|----|----|----|----|---------|---------|---------|---------|----------|----------|----------|----------|----------|-------|
| 1x152 double | 10 | 14 | 28 | 42 | 44.7225 | 61.7225 | 64.7225 | 94.7225 | 104.7225 | 144.7225 | 154.7225 | 169.7225 | 171.7225 | 177.7 |

$S_n$  table

In order to observe the relationship between  $M_n$  and  $n$  & between  $S_n$  and  $n$  clearly, I made two figures:



Obviously, the larger  $n$  is, the larger  $S_n$  is. In the second figure, as the number of samples increases, the value of  $M_n$  tends to flatten out and keeps "shaking" in a range.

Therefore, the study of  $M_n$  is to lead the subsequent study of sample variance, expectation and finally to estimate the population mean and analyze the population data.

The sample variance is defined as:

$$\sigma^2 = \frac{1}{n} \sum_{j=1}^n (x_j - M_n)^2$$

The expected value of the sample mean is given by:

$$E[M_n] = E\left[\frac{1}{n} \cdot \sum_{j=1}^n x_j\right] = \frac{1}{n} \sum_{j=1}^n E[x_j] = m$$

Sample mean is equal to  $E[x] = m$  on the average  $\rightarrow$  unbiased estimator for  $m$ .

The mean square error:

$$E[(M_n - m)^2] = E[(M_n - E[M_n])^2]$$

$$\therefore \text{VAR}[S_n] = \sum_{k=1}^n \text{VAR}[x_k] = n \text{VAR}[x_i] = n\sigma^2, \quad M_n = \frac{S_n}{n}$$

$$\therefore \text{VAR}[M_n] = \frac{1}{n^2} \text{VAR}[S_n] = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

I calculate the variance and the deviation for further study:

```
Sum_Sample_Square = 0;

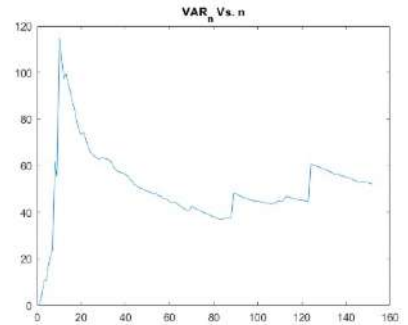
for n = 1:length(Average_TripMiles_Needed)
    Sum_Sample_Square = Sum_Sample_Square + (Average_TripMiles_Needed(n)-Mn(n))^2;
    Variance_1(n) = Sum_Sample_Square/n;
end

close all;
figure(4)
n = length(Average_TripMiles_Needed);
plot(1:n,Variance_1)
title('VAR_n Vs. n')

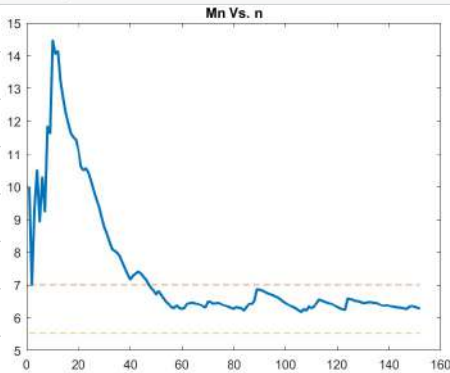
VAR_1_End = Variance_1(end)
Variance_2 = var(Average_TripMiles_Needed)
Target_Deviation = sqrt(Variance_2)/10

figure(5)

plot([1:n],Mn,'linewidth',2)
hold on
plot([1:n],ones(1,n)*(Mn(end)+Target_Deviation),'--')
plot([1:n],ones(1,n)*(Mn(end)-Target_Deviation),'--')
title('Mn Vs. n')
```



VAR\_1\_End = 52.1138  
 Variance\_2 = 54.3106  
Target\_Deviation = 0.7370



I got the deviation is 0.7370

And we can see the larger  $n$ , the  $Mn$  keeps "shaking" in that range.

b) As shown above,  $VAR[M_n] = \frac{\sigma^2}{n}$  when  $n$  increased, the sample mean approaches zero.

This implies that the probability that the sample mean is close to the true mean approaches one as  $n$  becomes very large.

This part is about deviation, population mean, average TRPMILES measurements... Consider the chebyshev inequality:

$$P(|M_n - E[M_n]| \geq \epsilon) \leq \frac{VAR[M_n]}{\epsilon^2}$$

The question is asking 'deviating the theoretical population  $\mu \leq \frac{6}{10}$ ', calculate the  $n$ . It can be wrote as

$$P(|M_n - \mu| < \frac{6}{10})$$

and consider the data

$$= 1 - \frac{6^2}{n\epsilon^2} = 1 - \frac{6^2}{\frac{n \cdot 6^2}{100}} = 1 - \frac{100}{n} = \frac{n-100}{n}$$

$$\therefore P \geq 0.9 \quad \therefore \frac{n-100}{n} \geq \frac{9}{10}$$

$$\therefore 10n - 1000 \geq 9n$$

$$n \geq 1000$$

As the result, 1000 measurements should be taken.

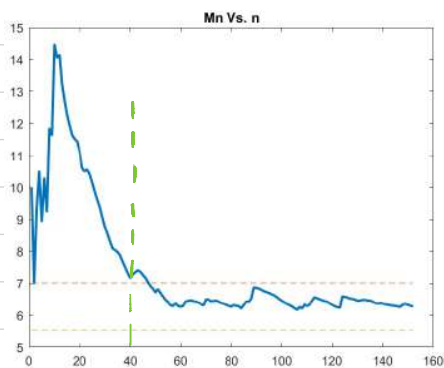
In the question 3a, we have only 152 measurement which is much less than 1000, it means the sample mean can not be used to evaluate the true mean. The department of transport should collect more data to inform regarding public transport development.

#### Question 4.

Explain the law of Large numbers to a government employee.

In this survey, for example, in question 3, we studied the relationship between mean of average TRPMILES and number of average TRPMILES measurements. A picture was obtained:

y-axis is mean, x-axis is number of measurements.



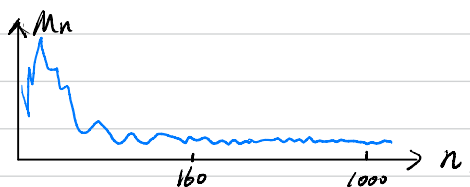
if we have only 40 measurements, (on the left of this green dotted line), It is obvious that the value of mean is unstable, the blue line fluctuates greatly in this region.

That means we can not evaluate a value of mean to represent

all the data.

However, if we add more data into study, as shown at the right hand side of the green dotted line, we see the blue line becomes more smooth and 'shaking' in a certain region (between the yellow dotted line and red dotted line). That means maybe we can evaluate a value of mean to represent all the data. But the 'mean' still not accurate enough.

Imagine if we have thousands of data, then the graph would change to



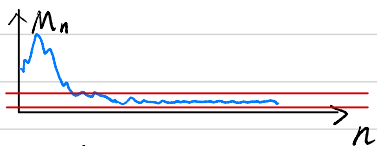
The smaller the jitter of the blue line, the more accurate the mean is, which is the theoretical population mean of the event occurring.

There are weak and strong versions of 'law of Large number'. Here is a very sample example to understand it:

- ① when the bell rang, the students in the classroom gradually quieted down and all of them keep in silent finally.
- ② when the bell rang, the students in the classroom gradually quieted down, most of them keep in silent finally except one person, he would suddenly scream sometimes.

① is strong law      ② is weak law

Or we can draw the  $M_n$  vs.  $n$  diagram again:



never beyond the borders  
→ strong law



will beyond the borders and come back,  
although with tiny probability  
→ weak law.